

Payments package

Package: com.example.androidexample.payments

Class: PremiumActivity

Methods I wrote and test coverage:

<u>Method name</u>	<u>Wrote test cases?</u>
onCreate(Bundle savedInstanceState)	✓
setupPremiumText()	✓
checkPremiumStatus()	✓ (indirect - exercised when the Activity launches)
isUserInPremiumList(JSONArray array, int currentUserId)	✗
startCheckout()	✓ (indirect - exercised via Upgrade button click test)
openStripeCheckout(String sessionUrl)	✗
getHeaders() (override inside JsonRequest in checkPremiumStatus)	✗

Test cases that cover these methods:

From PremiumActivityTest:

- premiumScreen_showsUI()
 - Launches PremiumActivity → executes onCreate()
 - Indirectly exercises setupPremiumText() and checkPremiumStatus() (Volley request created and queued).
- upgradeButton_isClickable()
 - Launches PremiumActivity → onCreate()
 - Clicks Upgrade button → executes startCheckout() (since isPremium defaults to false).
- returnHomeButton_isClickable()
 - Launches PremiumActivity → onCreate()
 - Clicks Return Home button (tests navigation click handler in onCreate()).

Tasks package

Package: com.example.androidexample.tasks

Class: Task

Methods I wrote and test coverage:

Method name	Wrote test cases?
Task(long taskId, String title, String description, String status, String dueDate)	✓
long getTaskId()	✓
String getTitle()	✓
String getDescription()	✓
String getStatus()	✓
String getDueDate()	✓

Tests exercising these methods:

From TaskTest:

- TaskTest.task_gettersReturnCorrectValues()
 - Creates a Task instance using the constructor and asserts all getter values, covering constructor + all getters.

Class: TaskAdapter

Methods I wrote and test coverage:

Method name	Wrote test cases?
TaskAdapter(Context context, List<Task> taskList, TaskManagerActivity activity)	✓
TaskViewHolder onCreateViewHolder(ViewGroup parent, int viewType)	✓
void onBindViewHolder(TaskViewHolder holder, int position)	✓
int getItemCount()	✓
TaskAdapter.TaskViewHolder(View itemView) (inner ViewHolder constructor)	✓

Tests exercising these methods:

From TaskAdapterTest:

- TaskAdapterTest.adapter_bindsDataWithoutCrashing_andCountsCorrectly()
 - Instantiates TaskAdapter → covers adapter constructor.
 - Calls onCreateViewHolder() and onBindViewHolder() → covers view inflation and data binding.
 - Calls getItemCount() and verifies it returns 1.
 - onCreateViewHolder() internally constructs TaskViewHolder, so the ViewHolder constructor is also covered.
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Class: TaskManagerActivity

Methods I wrote and test coverage:

Method name	Wrote test cases?
onCreate(Bundle savedInstanceState)	✓
void getTasks()	✓
void updateTask(long taskId, String title, String status, String dueDate)	✓
void deleteTask(long taskId)	✓
void showEditDialog(Task task)	✓
void showAddTaskDialog() (private helper)	✓
void addTask(String title, String description) (private helper)	✓

Tests exercising these methods

From TaskManagerActivityTest:

- onCreate
 - taskManagerActivity_launchesAndShowsRecyclerView()
 - taskManagerActivity_invalidUserId_finishesActivity()
 - recyclerView_hasCorrectLayoutManager()
 - recyclerView_hasAdapter()
 - allUIElements_areInitialized()
 - tvAddHint_existsInLayout()
 - getTasks_isCalledOnCreate() (verifies getTasks() ran on launch by checking adapter state)
- getTasks()

- Called inside onCreate() and explicitly in getTasks_handlesEmptyResponse().
- showAddTaskDialog() + addTask()
 - btnAddTask_opensAddTaskDialog()
 - addTaskDialog_hasCorrectElements()
 - addTaskDialog_cancelButton_dismissesDialog()
 - addTaskDialog_emptyFields_showsToast()
 - addTaskDialog_emptyTitle_showsToast()
 - addTaskDialog_emptyDescription_showsToast()
 - addTaskDialog_validInput_addsTask() → enters valid title/description and taps Add, which calls addTask() and then refreshes via getTasks().
- updateTask()
 - updateTask_canBeCalled() directly calls activity.updateTask(...).
 - editDialog_saveButton_updatesTask() opens showEditDialog(), edits title, taps Save, which triggers updateTask(...).
- deleteTask()
 - deleteTask_canBeCalled() directly calls activity.deleteTask(...).
- showEditDialog(Task)
 - showEditDialog_canBeCalled()
 - editDialog_hasCorrectElements()
 - editDialog_cancelButton_dismissesDialog()
 - editDialog_saveButton_updatesTask()

Tripplannar package

Package: com.example.androidexample.tripplanner

Class: AddEditTripEventActivity

Methods I wrote and test coverage:

Method name	Wrote test cases?
onCreate(@Nullable Bundle savedInstanceState)	✓
void openDateTimePicker(boolean isStart)	✓
String formatDisplay(String iso)	✓
void save()	✓

Tests exercising these methods

From AddEditTripEventActivityTest:

- onCreate
 - activity_launchesInNewMode() – verifies basic UI fields in “New Event” mode.
 - activity_launchesInEditMode() – verifies fields pre-populated from an existing TripEvent.
 - allFieldsInitialized() – asserts all views are non-null.
 - editMode_setsTitle() / newMode_setsTitle() – checks the title set in each mode.
 - editMode_loadsEventData() and editMode_loadsDates() – confirms edit mode loads data and formats dates correctly.
 - editMode_withNullPosition_doesNotCrash() – tests branch where position is null.
 - btnReturn_finishesActivity() – verifies the return button’s click listener (set in onCreate) calls finish().
- openDateTimePicker
 - edtStartsAt_opensDateTimePicker()
 - edtEndsAt_opensDateTimePicker()
 - openDateTimePicker_forStartsAt()
 - openDateTimePicker_forEndsAt()

These tests click the date/time fields and ensure the picker shows (and can be cancelled) without crashing.

- formatDisplay
 - formatDisplay_handlesValidIso() and editMode_loadsDates() – exercise the ISO → formatted date logic via the displayed text in edtStartsAt / edtEndsAt.
- save (validation + API branching)
 - save_missingName_showsError() – tests required name validation.
 - save_missingStartsAt_showsError() / save_missingEndsAt_showsError() – tests required start/end time validation.
 - save_invalidPosition_showsError() – tests invalid numeric parsing for position.
 - save_withValidData_callsCreateApi() – exercises the create branch (existingEvent == null).
 - save_withValidData_callsEditApi() – exercises the edit branch (existingEvent != null).
 - save_withEmptyOptionalFields_doesNotError() – tests that optional location/notes/position can be empty without errors.
 - save_withValidPosition_parsesCorrectly() – tests that a valid integer position is parsed and accepted.

Class: TripEventsActivity

Methods I wrote and test coverage:

Method name	Wrote test cases?
onCreate(@Nullable Bundle savedInstanceState)	✓
onResume()	✓ <i>(indirect – called by lifecycle)</i>
void loadEvents() (private helper)	✓ <i>(indirect – called by onCreate/onResume)</i>
void deleteEvent(TripEvent event) (private helper)	✗

Tests exercising these methods

From TripEventsActivityTest:

- onCreate + loadEvents
 - tripEventsActivity_launchesSuccessfully() – launches activity with valid tripId/userId, checks recyclerEvents and fabAddEvent are visible.
 - tripEventsActivity_allUIElementsDisplayed() – ensures recycler, FAB, and return button are displayed.
 - tripEventsActivity_missingTripId_finishesImmediately() / tripEventsActivity_missingUserId_finishesImmediately() – test validation branch where missing extras cause a toast + finish().
- onResume + loadEvents
 - onResume() is invoked as part of normal lifecycle whenever the tests launch the activity; loadEvents() is called from onResume, so event reloading is exercised indirectly.
- Return and FAB navigation
 - clickingReturn_finishesActivity() – verifies btnReturn’s click listener (set in onCreate) calls finish().
 - clickingFabAddEvent_opensAddEditTripEventActivity() – ensures FAB click starts AddEditTripEventActivity with correct intent.
- deleteEvent
 - No test directly triggers onItemLongClick/onDeleteClick, so deleteEvent(TripEvent event) currently has no dedicated test case.

Class: TripEventAdapter

Methods I wrote and test coverage:

Method name	Wrote test cases?
TripEventAdapter(OnItemClickListener listener)	✓
void setItems(List<TripEvent> events)	✓
EventViewHolder onCreateViewHolder(ViewGroup parent, int type)	✓
void onBindViewHolder(EventViewHolder holder, int position)	✓
int getItemCount()	✓
EventViewHolder(View itemView) (inner ViewHolder constructor)	✓
void bind(TripEvent event,.OnItemClickListener listener)	✓
String formatInstant(String iso) (private helper)	✓ (indirect via bind)

Tests exercising these methods

From TripEventAdapterTest:

- adapter_bindsEventWithoutCrash()
 - Creates a TripEventAdapter with a no-op OnItemClickListener → covers adapter constructor.
 - Calls setItems(Collections.singletonList(event)) → covers setItems.
 - Creates a FrameLayout parent and calls onCreateViewHolder(parent, 0) → covers onCreateViewHolder and EventViewHolder constructor.
 - Calls onBindViewHolder(adapter.onCreateViewHolder(parent, 0), 0) → covers onBindViewHolder, EventViewHolder.bind(...), and formatInstant(...).
 - Asserts getItemCount() equals 1.

Class: TripEvent

Methods I wrote and test coverage:

Method name	Wrote test cases?
static TripEvent fromJson(JSONObject)	✓

Tests exercising these methods

From TripEventTest:

- fromJson_parsingCorrectly()
 - Builds a full JSONObject with all fields and calls TripEvent.fromJson(obj).
 - Asserts every parsed field (IDs, name, times, location, notes, position, attachments, created/updated timestamps, and deleted flag).
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Class: TripEventPage

Methods I wrote and test coverage:

Method name	Wrote test cases?
static TripEventPage fromJson(JSONObject)	✓

Tests exercising these methods

From TripEventPageTest:

- fromJson_parsingPageCorrectly()
 - Builds a JSON page object with content, number, size, totalPages, totalElements.
 - Calls TripEventPage.fromJson(root) and asserts:
 - content list size and first TripEvent.id
 - number, size, totalPages, totalElements values.
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Class: TripEventApi

Methods I wrote and test coverage:

Method name	Wrote test cases?
TripEventApi(Context context)	✓
void listEvents(long userId, long tripId, String fromIso, String toIso, int page, int size, PageCallback callback)	✓
void createEvent(long userId, long tripId, String name, String startsAtIso, String endsAtIso, String location, String notes, Integer position, EventCallback cb)	✓
void editEvent(long userId, long tripId, long eventId, String name, String startsAtIso, String endsAtIso, String location, String notes, Integer position, EventCallback cb)	✓

<code>void deleteEvent(long userId, long tripId, long eventId, SimpleCallback callback)</code>	✓
<code>Map<String,String> buildHeaders(long userId) (private helper)</code>	✓ <i>(indirect via getHeaders())</i>
<code>String parseVolleyError(VolleyError error) (private helper)</code>	✓ <i>(indirect when errors occur)</i>

Tests exercising these methods

From TripEventApiTest:

- Instantiation + public API methods
 - `api_instantiatesCorrectly()` – constructs `TripEventApi`, ensuring constructor works.
 - `listEvents_executesWithoutCrash()` – calls `listEvents(...)` with a dummy callback (exercises URL build, request creation, and header building).
 - `createEvent_executesWithoutCrash()` – calls `createEvent(...)` with all fields set (exercises JSON body creation and request queue submission).
 - `editEvent_executesWithoutCrash()` – calls `editEvent(...)` similarly.
 - `deleteEvent_executesWithoutCrash()` – calls `deleteEvent(...)`.
- Surrogate tests for URL/JSON behavior (supporting the same logic used inside methods):
 - `urlContainingTripId_buildsCorrectly()` and `urlWithOptionalParams_buildsCorrectly()` – verify URL patterns with trip IDs and optional from/to params.
 - `createEvent_jsonHasRequiredFields()` and `createEvent_optionalFieldsIncluded()` – verify expected JSON keys/values.
 - `editEvent_handlesNullFields()` – verifies behavior when optional fields are absent.
- Surrogate tests for error handling
 - `parseVolleyError_messageExists()`, `parseVolleyError_nullMessage()`, and `parseVolleyError_httpStatus()` – validate how `VolleyError` carries message/status, mirroring conditions handled by `parseVolleyError(...)`.

(Because `buildHeaders` and `parseVolleyError` are private, they are exercised indirectly through the request objects and volley callbacks, and conceptually covered by these surrogate tests.)

Class: `ApiClient`

Methods I wrote and test coverage:

Method name	Wrote test cases?
ApiClient(Context context) (private constructor)	✓ (indirect via getInstance)
static synchronized ApiClient getInstance(Context)	✓
RequestQueue getRequestQueue()	✓
<T> void addToRequestQueue(Request<T> req)	✗

Tests exercising these methods

From ApiClientTest:

- apiClient_singletonWorks()
 - Calls ApiClient.getInstance(...) twice and asserts both references are the same singleton instance → covers constructor + getInstance.
 - Calls c1.getRequestQueue() and asserts it is non-null → covers getRequestQueue.

Note: addToRequestQueue(...) is used by TripEventApi, but there is no test that directly asserts behavior specific to this method, so it is marked as no dedicated test case.

MuhammadSystemTest - Mini Assignment

Class: LoginActivity

Methods I wrote and test coverage (from MuhammadSystemTest):

Method name	Wrote test cases?
onCreate(Bundle savedInstanceState)	✓
loginUser(String email, String password)	✓ (via login_typing button click)
fetchUsernameById(int userId)	✗
promptForUsernameThenContinue(int userId)	✗
saveSession(int userId, String username)	✗
goFriends(int userId, String username)	✗
isValidEmail(String s)	✓ (called in btnLogin click validation)

Tests touching these:

- login_showsFields() → launches LoginActivity ⇒ runs onCreate.
 - login_typing() → launches LoginActivity, types email/password, clicks Login ⇒ runs onCreate, then isValidEmail(...) and loginUser(...).
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Class: HomeActivity

Methods I wrote and test coverage (from MuhammadSystemTest):

Method name	Wrote test cases?
onCreate(Bundle savedInstanceState)	✓
onBackPressed()	✗
loadUserProfile(int userId)	✓ (called inside onCreate when userId is valid)
onPause()	✓ * (via Activity lifecycle, no explicit assertion)
onResume()	✓ * (via Activity lifecycle, no explicit assertion)
onStop()	✓ * (when ActivityScenario closes, no explicit assertion)
onDestroy()	✓ * (when ActivityScenario closes, no explicit assertion)

***Lifecycle methods are invoked by ActivityScenario, but you don't have explicit assertions on them; they're still executed during the system test run.**

Test touching these:

- home_showsUsernameAndDrawer() → uses launchHome() (starts HomeActivity with userId extra) ⇒ runs onCreate(), which calls loadUserProfile(userId).
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Class: TripListActivity

Methods I wrote and test coverage (from MuhammadSystemTest):

Method name	Wrote test cases?
onCreate(@Nullable Bundle savedInstanceState)	✓
onResume()	✓ * (Activity lifecycle after launch)

loadData()	✓ (called from onCreate / onResume)
showUpsertDialog(boolean isCreate, @Nullable TripDTO existing)	✗
showInviteDialog(TripDTO trip)	✗
confirmDelete(TripDTO t)	✗

Tests touching these:

- tripList_showsUI() → uses launchTripList() ⇒ runs onCreate() and initial loadData(), then checks trips UI.
 - tripList_openInvites() → launches TripListActivity, then clicks View Invites button ⇒ still exercises onCreate()/loadData().
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Class: TripInvitesActivity

Methods I wrote and test coverage (from MuhammadSystemTest):

Method name	Wrote test cases?
onCreate(@Nullable Bundle savedInstanceState)	✓
onResume()	✓* (lifecycle)
loadInvites()	✓ (explicitly called from onCreate)

Tests touching these:

- tripList_openInvites() → clicking View Invites on TripListActivity starts TripInvitesActivity ⇒ runs onCreate() and loadInvites().
 - invites_showList() → directly uses launchInvites() ⇒ runs onCreate() and loadInvites(), then checks recyclerInvites.
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Class: TripChatActivity

Methods I wrote and test coverage (from MuhammadSystemTest):

Method name	Wrote test cases?
onCreate(Bundle savedInstanceState)	✓
onEdit(TripMessageResponseDTO msg)	✗

onDelete(TripMessageResponseDTO msg)	X
onReact(TripMessageResponseDTO msg)	X
onUnreact(TripMessageResponseDTO msg, String emoji)	X

Test touching these:

- chat_sendMessage() → uses launchChat() ⇒ runs onCreate() and sets up RecyclerView, ViewModel, and send button; then types a message and clicks Send, which exercises the send-message logic inside onCreate (calls vm.send(...)).

Class: TripEventsActivity

(Trip planner screen – this is also part of the Tripplanner package, but here we're just showing what MuhammadSystemTest covers.)

Methods I wrote and test coverage (from MuhammadSystemTest):

Method name	Wrote test cases?
onCreate(@Nullable Bundle savedInstanceState)	✓
onResume()	✓* (lifecycle)
loadEvents()	✓ (called from onCreate / onResume)
deleteEvent(TripEvent event)	X

Test touching these:

- planner_showsUI() → uses launchPlanner() ⇒ runs TripEventsActivity.onCreate(), which sets up RecyclerView, FAB, and calls loadEvents(), then the test asserts recyclerEvents and fabAddEvent are visible.

Class: MuhammadSystemTest

Helper methods in MuhammadSystemTest:

Method name	Purpose
launchHome()	Starts HomeActivity with userId 52
launchTripList()	Starts TripListActivity with userId 1
launchChat()	Starts TripChatActivity for trip 123

launchInvites()	Starts TripInvitesActivity for user 1
launchPlanner()	Starts TripEventsActivity for trip 123

Test cases in MuhammadSystemTest:

Test method name	What it verifies
login_showsFields()	Login screen fields visible
login_typing()	Can type email/password and tap Login
home_showsUsernameAndDrawer() ()	Home drawer and username view visible
tripList_showsUI()	Trips list screen basic UI visible
tripList_openInvites()	Trips → View Invites navigation works
invites_showList()	TripInvites list RecyclerView is displayed
chat_sendMessage()	Chat screen can type and send a message
planner_showsUI()	Trip planner events screen basic UI visible